SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS: **Total Marks: 20**

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.

4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I D.Suryakiran/192325057(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

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CO2 - AT3 - Class Test

1. Machine learning Vs Deeplearning :-

Conceptual definition:-

- Machine learning [ML] : It is a branch of AI where algorithms learn patterns from data and make predictions / decisions without being explicitly program for each rule.
- Deep Learning (DL) : It is a subset of ML that uses multi-layered neutral networks to automatically learn hierarchial feature representations from raw data.

Comparision table :-

Aspect	Machine Learning	Deep Learning
Architeture	Shallows models :linear / logistic regression, SVM, decision tree.	Deep neural network like CNN, RNNs, Transformers
Data requirements	Works well with small to medium data sets, Structured tabular data often sufficient.	Needs large-scale datasets, excels on unstructed data
feature extraction	Manual: domain experts design extract / features	Automatic: learns features directly from raw input via layered representations.
Training & hardware	faster training : typically runs on CPU's less compute-intensive	slower training : usually needs GPUS / TPU's high compute and memory
Interpretability	Generally more Interpretable	often a " black box" hard to interpret
Accuracy	Good when features are well-designed and data is limited	Higher accuracy on complex tasks if enough data.

Real-world applications:

*Machine Learning applications

1. Spam detection in email : To classify emails are spam are not
2. Stock price prediction: historical price and indicate data.
3. Recommendation systems: suggests products, price and movie
4. fraud detection in transcations: suspicous banking
5. Predective analysis in finance/health : disease risk, fore casts.

* Deep learning applications

1. Image classification : medical imaging , scanning.
2. Object and face detection: Locates and recognizes faces) objects
3. Speech recognition (ASR): language translation.
4. Natural language processing (NLP): power translation, sentiment analysis.
5. ~~Autonomous~~ veechicles: lane detection and navigation by camera.

∴ Therefore ML and Deep learning are complementary pillar of ~~modern AI~~. AIML prodivdes efficient interpret solutions

* The choice between them depends upon data avaliability, task, complexicity.

2. Representation Learning : Effect of Network width and Depth.

Representation Learning:

Representation learning is the process by which a model automatically discovers useful representations features from raw data that make down stream tasks.

Width:-

Number of units/neurons per layer in neurons

Depth:-

Number of layer in neurons.

Influence of learning capability:

- Depth enables hierarchical composition of features [Simple → complex]
- Improving expressivity for complex functions. [edges → shapes → objects in vision].
- Width increases capacity within a layer, allowing more parallel feature dectors.
- Better approximation at a given depth.
- Very wide or very deep networks both increase model capacit
- deeper nets favour compositional hierarchies.
- wider nets favour richer-single layer representations.

Computational complexicity:

- Depth increases sequential operations (more layers to pass through).
- Raising training time and memory of backpropagation.
- Width increases per layers computation (more weights, more flops)
- overall complexicity grows the total parameters.

Model performance:-

- Properly increasing depth/width can improve accuracy by better.
- approximately complex functions, but risks, overfitting if data is limited.
- The depth to width ratio controls effective model complexicity and durations.
- There often an optimal aspect ratio for best generalizati
- Too much depth without normalization/residuals can hurt training.
- To much width can waste comput and overfit.

8. ReLU VS Leaky ReLU VS ELU

Let x be the input to the activation

Mathematical definitions:-

- ReLU (Rectified Linear unit)

$$ReLU(x) = max(0, x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases}$$

- Leaky ReLU (LReLU)

with small slope $\alpha \in (0, 1)$, eg 0.01)

$$LReLU(x) = \begin{cases} \alpha x, & x \leq 0 \\ x, & x > 0 \end{cases}$$

- ELU (Exponential Linear Unit)

(with parameter $\alpha > 0$)

$$ELU(x) = \begin{cases} \alpha(e^{x} - 1), & x \leq 0 \\ x, & x > 0 \end{cases}$$

Behaviour and derivatives:

- ReLU

Derivative : 1 for $x > 0$,

0 for $x < 0$,

creates sparse activations (many zeros.

- Leaky ReLU
- Derative : 1 for $x > 0$

α for $x < 0$

Always has a small gradient for negative inputs.

• ELU:

• Derivative : 1 for $x > 0$

Elu(x) + α for $x < 0$ (smooth near zero).

Negative side is smooth and saturates to $-\alpha$ as $x \to -\infty$

advantages/disadvantages:-

Function	Advantages	disadvantages.
ReLU	- Very cheap to compute - Non-saturating for $x > 0$ + fast	- Dying ReLU: - Unbounded positive output.
Leaky ReLU	- Mitigates dying ReLU by allowing - still computationall	- slope α is fixed heuristic tasks
ELU	- smooth continous derivatives → better → Mean activation closer to zero	- More expensive due to exponential computation - can still Saturate for large.

4. Unsupervised learning in neural networks.

RBMS and Auto encoders:-

Unsupervised learning:

Unsupervised learning learns trains models without labeled data, targets.

RBM's [Restricted Boltzman Machines]

Archiecture:-

- Bipartite undirected graphical model with a visible layer v and a hidden layer h;
- Defines a joint probability distribution P(v,h) via energy model.

working principle

- Learns weights and baises so that model assign high probability to observe data.
- Hidden units capture latent features;

Applications

- Feature learning / pretraing for deep networks
- Pattern recognition,
- radar target recognition

Autoencoders:-

Architecture:-

- Directed neural network with an encoder mapping map input x to latent code $z = f_\theta(x)$

decoder reconstructing $\hat{x} = g_\phi(z)$

working principle:-

- Training to minimize reconstruction error.
- sparse, denoising, variational.
- The bottleneck forces the network to learn

applications:-

1) Dimensionality reduction
2) denoising
3) anomaly detection
4) pretraining for downstream tasks